

DATA STRUCTURE

LAB PROGRAMS

9. a) Write a C program for implementing array and perform insertion .

```
#include <stdio.h>

int main()
{
    int arr[100], n, i, pos, value;
    printf("Enter number of elements: ");
    scanf("%d", &n);
    printf("Enter array elements:\n");
    for(i = 0; i < n; i++)
    {
        scanf("%d", &arr[i]);
    }
    printf("Enter position for insertion: ");
    scanf("%d", &pos);
    printf("Enter value to insert: ");
    scanf("%d", &value);
    for(i = n; i >= pos; i--)
    {
        arr[i] = arr[i - 1];
    }
    arr[pos - 1] = value;
    n++;
    printf("Array after insertion:\n");
    for(i = 0; i < n; i++)
```

```

    {
        printf("%d ", arr[i]);
    }
    return 0;
}

```

Output

```

Enter number of elements: 5
Enter array elements:
10 20 25 30 40
Enter position for insertion: 4
Enter value to insert: 35
Array after insertion:
10 20 25 35 30 40

=== Code Execution Successful ===

```

b) Write a C program for implementing array and perform deletion .

```

#include <stdio.h>

int main()
{
    int arr[100], n, i, pos;

    printf("Enter number of elements: ");
    scanf("%d", &n);
    printf("Enter array elements:\n");
    for(i = 0; i < n; i++)

```

```

{
    scanf("%d", &arr[i]);
}
printf("Enter position to delete: ");
scanf("%d", &pos);
for(i = pos - 1; i < n - 1; i++)
{
    arr[i] = arr[i + 1];
}
n--;
printf("Array after deletion:\n");
for(i = 0; i < n; i++)
{
    printf("%d ", arr[i]);
}
return 0;
}

```

Output

Enter number of elements: 5

Enter array elements:

10 20 30 40 45

Enter position to delete: 3

Array after deletion:

10 20 40 45

=== Code Execution Successful ===

c) Write a C program for implementing array and perform updation .

```
#include <stdio.h>

int main()
{
    int arr[100], n, i, pos, value;
    printf("Enter number of elements: ");
    scanf("%d", &n);
    printf("Enter array elements:\n");
    for(i = 0; i < n; i++)
    {
        scanf("%d", &arr[i]);
    }
    printf("Enter position to update: ");
    scanf("%d", &pos);
    printf("Enter new value: ");
    scanf("%d", &value);
    arr[pos - 1] = value;
    printf("Array after updation:\n");
    for(i = 0; i < n; i++)
    {
        printf("%d ", arr[i]);
    }
    return 0;
}
```

Output

Enter number of elements: 5

Enter array elements:

10 20 25 30 40

Enter position to update: 3

Enter new value: 27

Array after updation:

10 20 27 30 40

=== Code Execution Successful ===